



Session 2: RVSM & Navigation

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RNP/RNAV

The ability to conduct Required Navigation Performance (RNP) and Area Navigation (RNAV) operations is rapidly becoming a standard requirement for commercial air carriers.

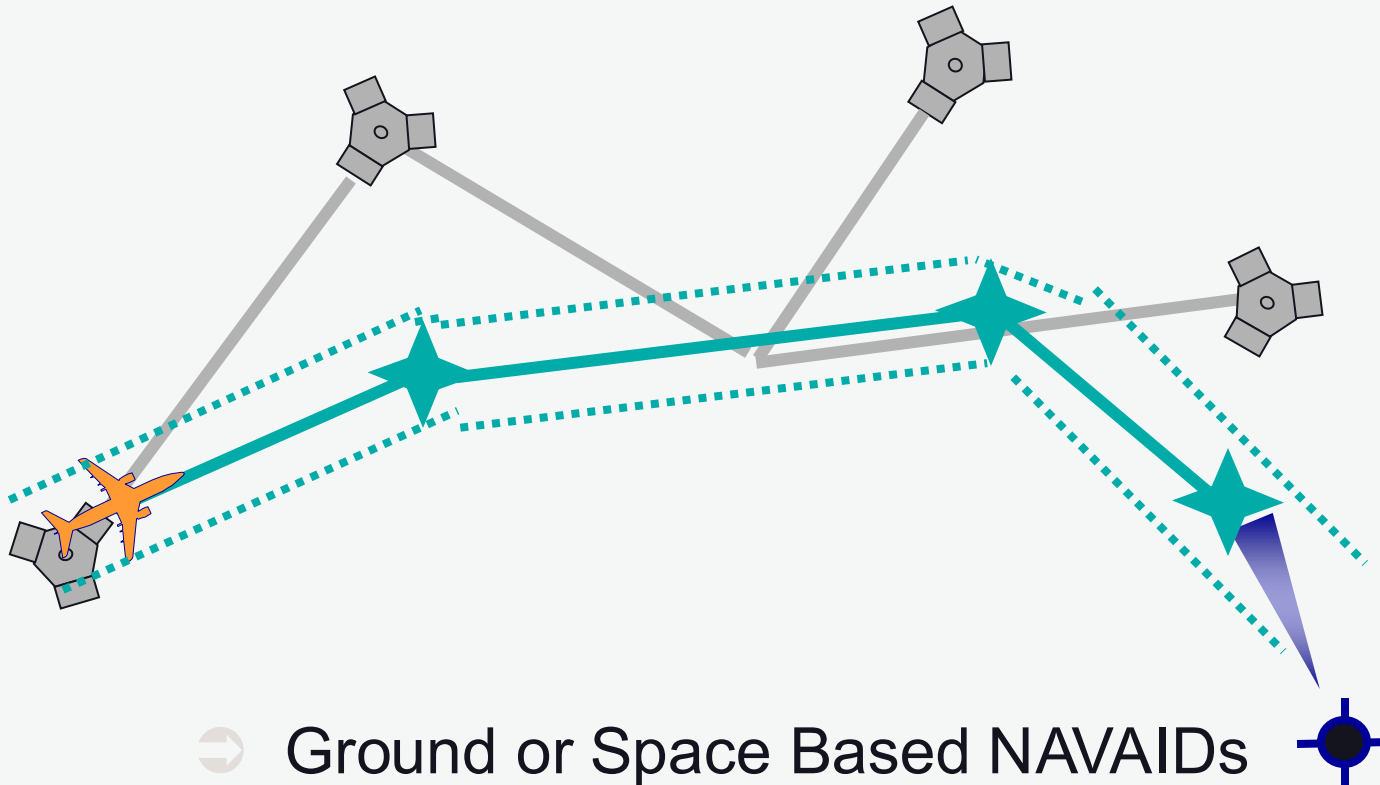
RNP capability is necessary to gain access to large areas of airspace around the world.

RNAV capability increases navigation flexibility and permits more efficient operations thus offering economic benefits to airlines.

Difference Between RNAV and RNP

- RNAV is a method of navigation that enables aircraft to fly on any desired flight path within the coverage of referenced NAVAIDS or within the limits of the capability of self-contained systems, or a combination of these capabilities
- Routes and procedures using RNAV provide improved access and flexibility through point-to-point navigation and are not restricted to the location of ground-based NAVAIDS
- The overall safety of the RNAV operation is achieved through a combined use of aircraft navigation accuracy, air traffic control intervention (via radar monitoring, automatic dependent surveillance (ADS), multilateration, communications) and/or route separation.
- The concept of PBN relies on the use of an area navigation system

Area Navigation (RNAV)



➡ Ground or Space Based NAVAIDs

➡ Aircraft Fly Waypoints

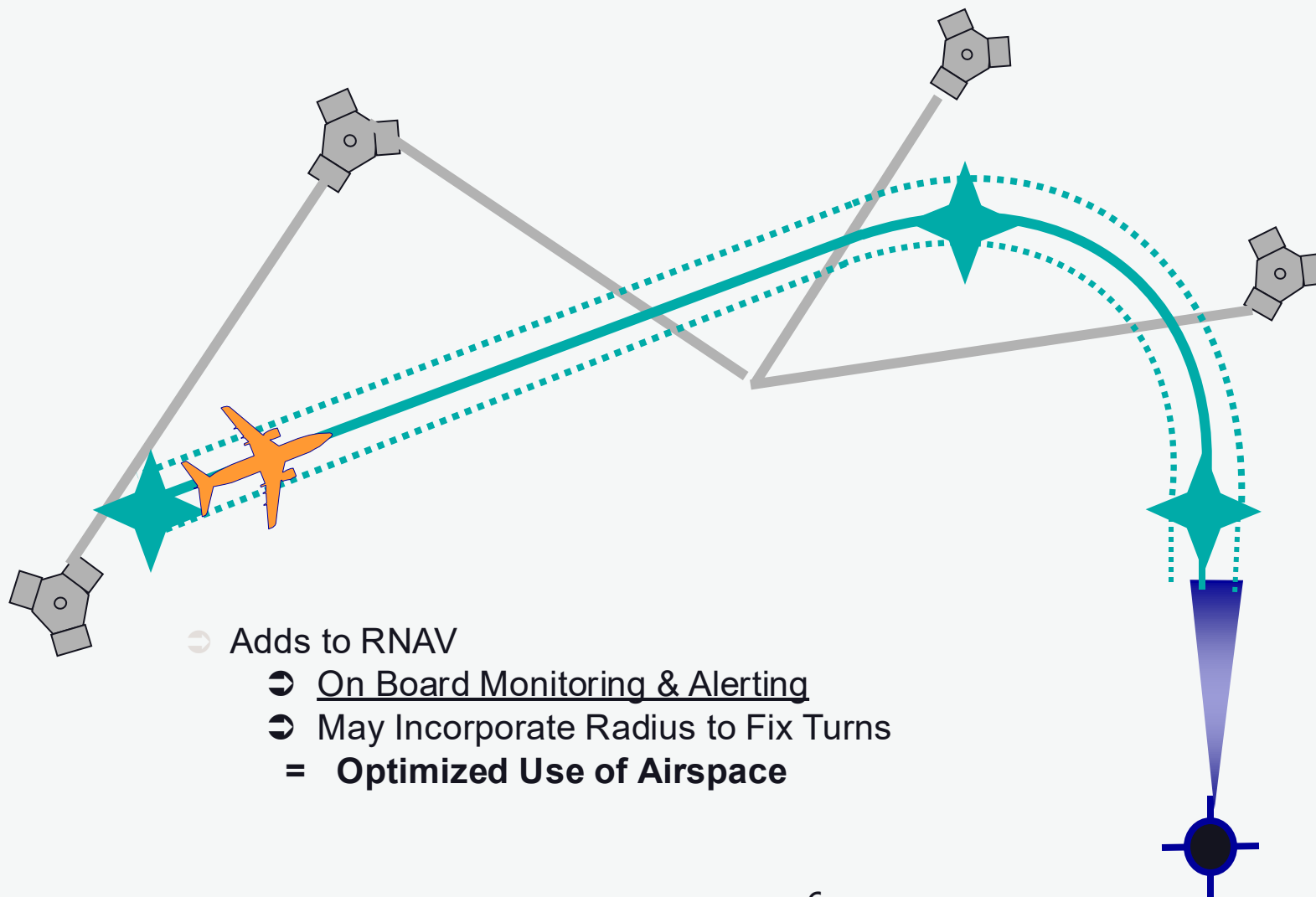
➡ Protected Area Constant (“Linear”)

= **Increased Design Flexibility**

Difference Between RNAV and RNP

- RNP is RNAV with on-board navigation monitoring and alerting
- RNP is a statement of navigation performance necessary for operation within a defined airspace
- A critical component of RNP is the *ability of the aircraft navigation system to monitor its achieved navigation performance, and to identify for the pilot whether the operational requirement is, or is not being met during an operation*
- This on-board monitoring and alerting capability therefore allows a lessened reliance on air traffic control intervention (via radar monitoring, automatic dependent surveillance (ADS), multilateration, communications), and/or route separation to achieve the overall safety of the operation
- RNP capability of the aircraft is a major component in determining the separation criteria to ensure that the overall containment of the operation is met

Required Navigation Performance(RNP)



Concept Of RNP

- RNP as a concept applies to navigation performance within an airspace. As such, it affects both the airspace and the aircraft within that airspace. RNP is intended to characterize an airspace through a statement of the navigation performance accuracy to be achieved within that airspace. This is known as the RNP type. RNP type is based on the expectation that all aircraft within the airspace will achieve the specified level of navigation accuracy 95% of the time.
- The development of the RNP concept recognizes that current aircraft navigation systems are capable of achieving a predictable level of navigation performance accuracy. Based on this capability, more efficient use of airspace can be realized. States (Governments) determine the RNP type based on a number of interrelated variables.

RNP TYPES

RNP Type	Accuracy
RNP0.1	$\pm 0.1\text{nm}$ (Approach Area for Landing)
RNP0.3	$\pm 0.3\text{nm}$ (Approach Area for Landing)
RNP1	$\pm 1.0\text{nm}$ (Terminal Area for ARR/DEP)
RNP4	$\pm 4.0\text{nm}$
RNP5	$\pm 5.0\text{nm}$
RNP10	$\pm 10.0\text{nm}$
RNP12.6	$\pm 12.6\text{nm}$
RNP20	$\pm 20.0\text{nm}$

RNP RNAV Functionality

- Regardless of the level of automation and the accuracy of the navigation system ,flight crews must be able to maintain situational awareness.
- This included awareness of what the automation is doing at any given time. Therefore, full RNP RNAV functionality must include a flight crew interface through the Flight Management Control Display Unit (FM CDU).
- The Primary Flight Displays and Crew Alerting System must reflect navigation, flight planning, warning ,functions, and database RNP values.

Operational Consideration

RNP operation is certified in the Airplane Flight Manual which will stipulate any limitations that might be imposed on RNP operations, for example, RNP without GPS .

Flight Planning & Preflight procedures

- Confirm that the aircraft is qualified for RNP operations 'R' is in block 10 of the ICAO flight plan
- Review MEL items to ensure RNP-10 restriction apply.

B-RNAV Operations.

B-RNAV stands for basic area navigation and is European RNP-5 operations. Aircraft flying within designated BRNAV airspace must use self contained area navigation equipment that automatically determines the aircraft position by one or more sensors.

Training Requirements

FLIGHT CREW TRAINING :

- Initial and recurrent training should ensure that flight crews :
 - Have a general knowledge of the application of RNP ;
 - Have a thorough understanding of the equipment
 - Are aware of its limitations
 - Have been trained in the operating procedures and safeguards necessary to obtain optimum efficiency and maintenance of required navigational accuracy;
 - Are in current practice and have received recent training on the equipment.
 - Are conversant with the contingency procedures in case of equipment failure.

Training Requirements.

DISPATCHER TRAINING.

- Dispatchers should receive training on RNP operations that is similar to that given to flight crews. They should:
 - Have a general knowledge of the application of RNP;
 - Be aware of its limitations;
 - Understand the effect of MEL, items on flight planning; and:
 - Understand the requirements for contingency operations and their effect on fuel planning.

Performance Based Navigation (PBN)

performance-based navigation (PBN) represents a shift from sensor-based to performance-based navigation. PBN specifies that aircraft RNP and RNAV system performance requirements be defined in terms of accuracy, integrity, availability, continuity and functionality required for the proposed operations in the context of a particular airspace, when supported by the appropriate navigation infrastructure.

GF Authorization

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Given the transition status of implementing PBN over Gulf Air's network and the existence of inconsistent navigation specifications, Gulf Air crew are authorized to fly RNAV procedures only as described in the table below.

Navigation Specification On Chart	Aircraft Systems and Crew Procedures required for Airbus	Aircraft Systems and Crew Procedures required for Boeing
RNAV-2, RNP-1, RNAV 1, P-RNAV	FCOM PRO-SPO-51 P-RNAV/RNP-1	FCTM 1.30 – 1.34
RNP-5, RNAV 5, B-RNAV	FCOM PRO-SPO-51 BRNAV	FCTM 1.30 – 1.34
RNP-10	FCOM PRO-SPO-51 RNP 10, RNP	FCTM 1.30 – 1.34
RNP-4	FCOM PRO-SPO-51 RNP-4	FCTM 1.30 – 1.34
RNAV SID/STAR/TRANSITION with no ICAO RNAV/RNP or NAVAID or Aircraft Equipment specification	FCOM PRO-SPO-51 BRNAV	FCTM 1.30 – 1.34
RNAV SID/STAR/TRANSITION with no ICAO RNAV/RNP specification, but with NAVAID or aircraft equipment specified e.g. VOR/DME, DME/DME, FMS, GPS	FCOM PRO-SPO-51 BRNAV	FCTM 1.30 – 1.34
RNAV / RNP / GPS Approach	FCOM PRO-NOR-SOP	FCTM 5.27 – 5.58, QRH MAN 2.3 - 2.6

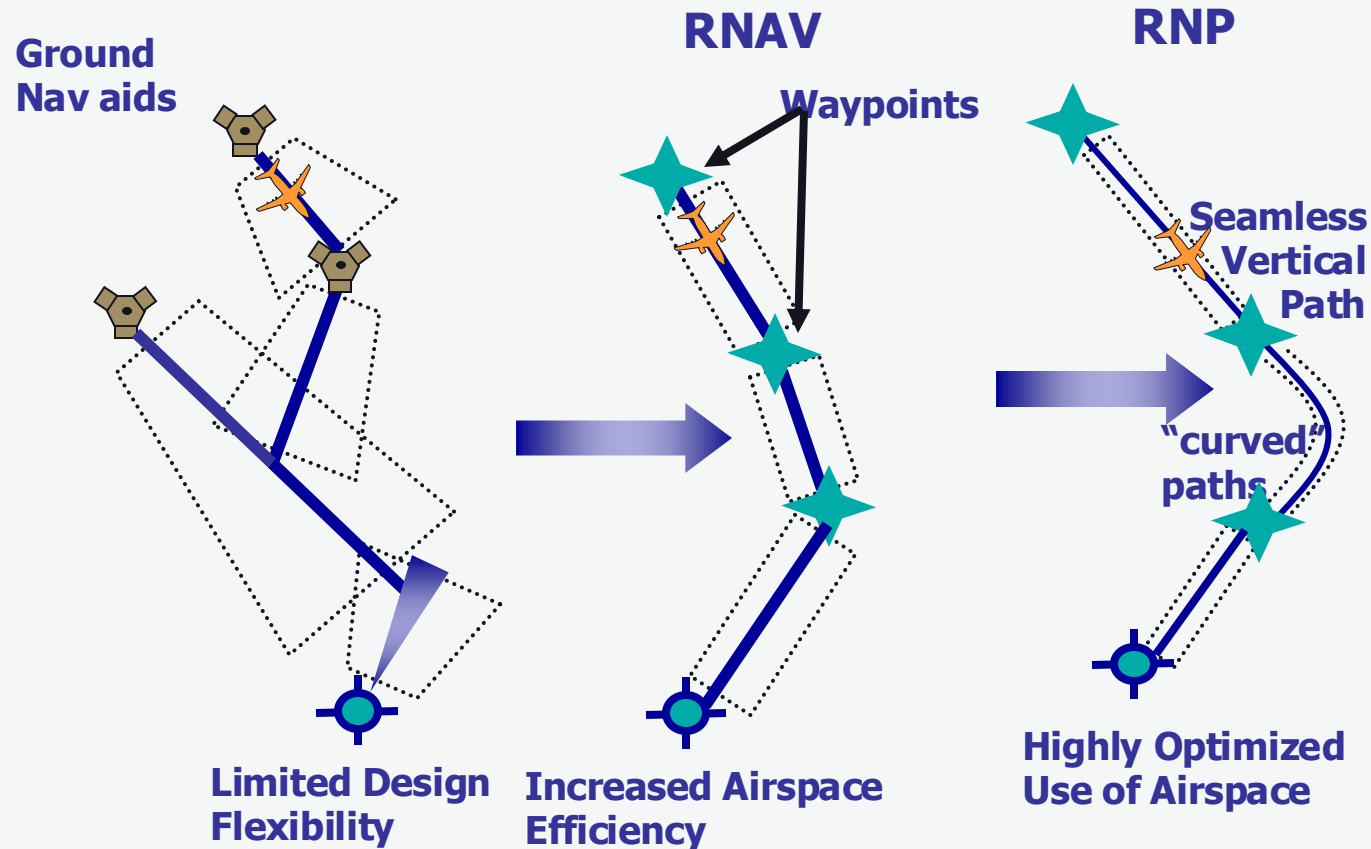
Note: In some Navigation application, charts will state GNSS (GPS) as being required for the RNAV procedure, due to insufficient ground NAVAID coverage. In these cases, "GPS/GPS PRIMARY" is mandatory.

Advantages of PBN

- Flight safety will be improved through harmonized Navigation Applications.
- Safety (precise trajectory management, CDA)
- Predictability
- Efficiency (min air distance / optimum aircraft determined profile)
- Minimise environmental impact (CO₂ NO_x & noise)
- Maximise Capacity utilisation (aerodrome & airspace)
- High aircraft utilisation (sectors per day)
- Schedule reliability
- Cost effective investment
- Minimised quantity of CTA and optimise design

Summary of the evolution

The use of satellite technology has allowed the aviation industry to move away from its dependence upon ground based navigation systems and gain more airspace



Minimum Requirement Equipment.

Please refer to MEL PBN section.

ACAS II(TCAS)

WHAT IS ACAS(TCAS):

- ACAS (Airborne Collision Avoidance System) is also known as TCAS (Traffic Alert and Collision Avoidance System).
- It is a transponder based interrogation system capable of displaying conflicting traffic and providing resolution advice.
- Provides an extra level of protection against mid-air collision.
- ACAS is backup to the ATC system ,it provides safe separation between aircraft are predicated to be on collision trajectories while minimizing ATC clearance deviations or excursions.

How It Works

- TCAS surveys the airspace around the host aircraft by actively interrogating the transponders on other aircraft and listening for their replies.
- TCAS recognizes Mode A, Mode C and Mode S replies. The system computes the range, relative A bearing, relative altitude (if the target aircraft altitude reporting is active), and A rate of closure.
- Using this information, inputs from the host aircraft systems and knowledge about the performance capabilities of the host, the computer predicts the flight paths of both the host and target aircraft, estimates the closest point of approach, and determines if a potential conflict exists.
- If a conflict exists, the system provides guidance for the optimum VERTICAL avoidance maneuver ;

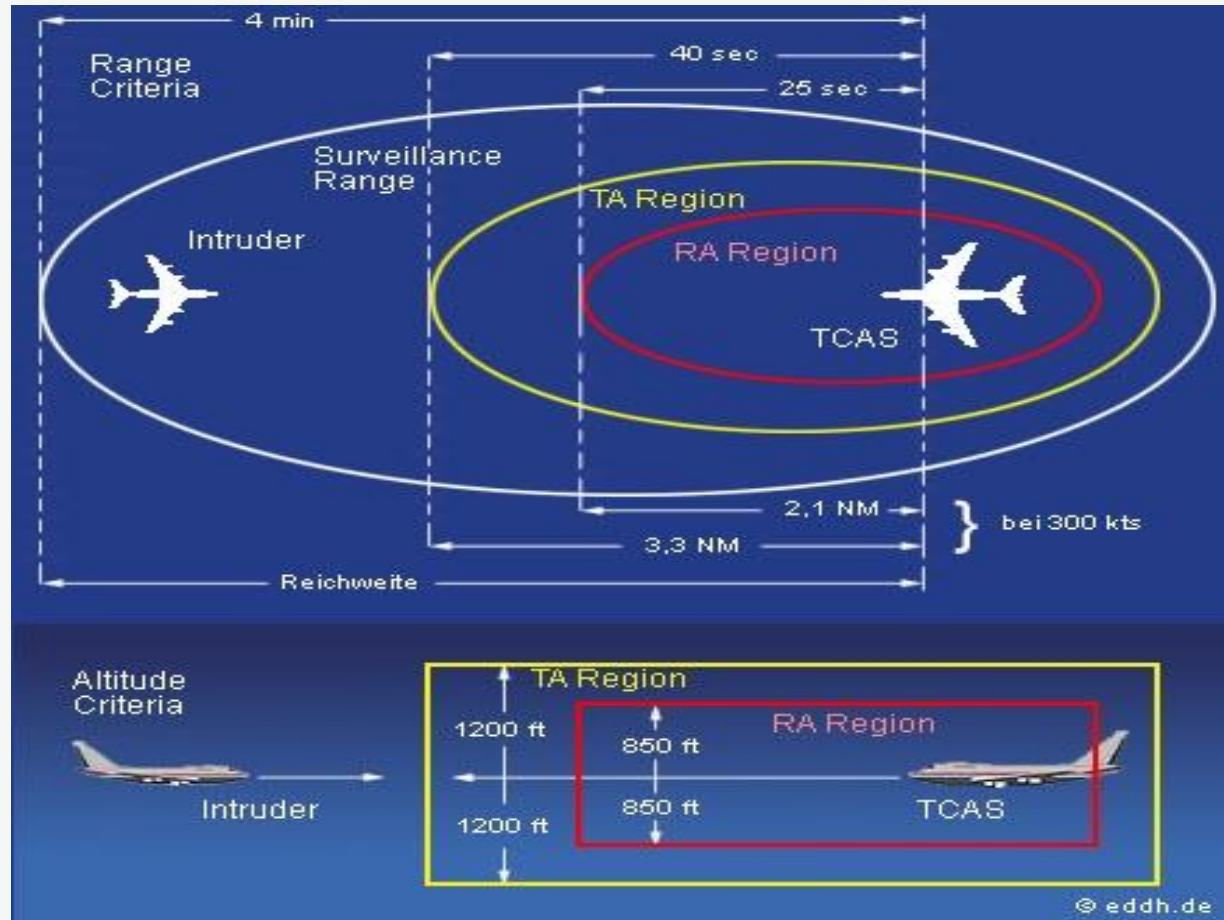
ACAS II

- ACAS II is capable of providing resolution advisories (RA) anytime an aircraft enters the RA range and altitude limits of the ACAS.
- The RA will be a combination of aural and visual cues.
- Traffic with Mode A transponders do not allow the ACAS to compute an RA.
- Traffic with Mode C transponders allow the ACAS to determine and display a RA.
- Traffic with Mode S transponders allow the two aircraft to communicate via data link and provide complimentary RA's. EX: one aircraft climbs while the other descends.

Advisories

- Traffic Advisory (TA)- help the pilots in the visual search for the intruder aircraft, and alert them to be ready for a potential resolution advisory.
- Resolution Advisory (RA)- avoidance maneuvers recommended to the pilot; when the intruder aircraft is Mode S equipped both systems will co-ordinate their RA's through data link.
- Targets are displayed on the ACAS screen as different symbols which depict level of threat.
- The visual cues will be accompanied by an aural warning. EX: TA "Traffic, Traffic" or RA "Climb, Climb"
- Relative altitude will be displayed in conjunction with a trend arrow.
- Relative altitude is the targets altitude in relation to the aircraft. EX: -03↑ depicts a target 300 feet below and climbing at 600fpm or more.
- If the target is climbing or descending at less than 600fpm the arrow will not appear.
- ACAS is incapable of displaying aircraft without transponders.

Protection Thresholds



TYPICAL ACAS II DISPLAY



Figure 9: TCAS traffic display example – IVSI combined with TCAS traffic display.



Figure 10: TCAS traffic display example - Electronic Flight Instrument System (EFIS).

PILOT/CONTROLLER ACTION

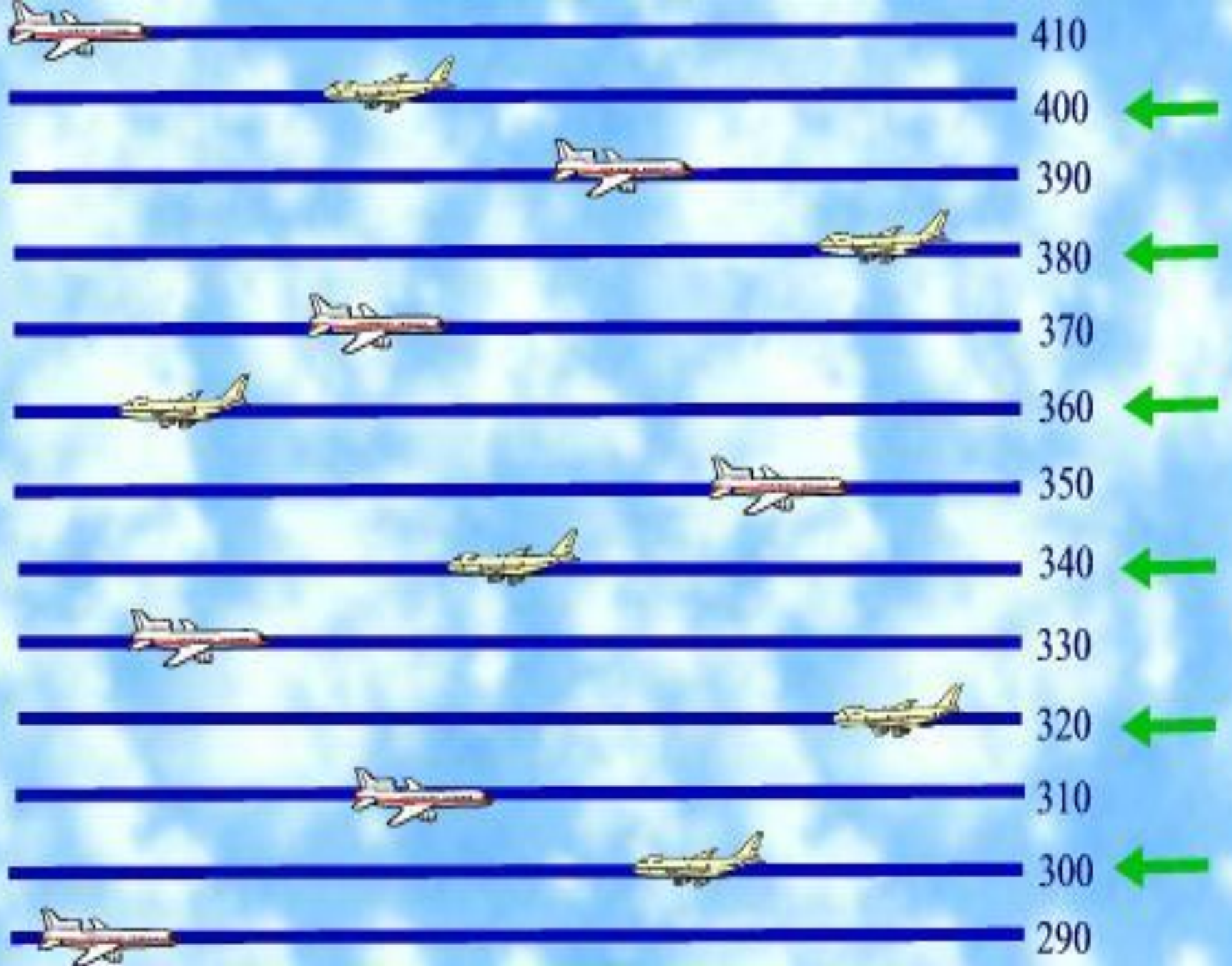
- Pilots shall not maneuver their aircraft in response to TA's only.
- In the event of an RA to alter flight path the alteration should be the minimum required and ATC must be notified as soon as possible.
- Pilots who deviate from an ATC instruction or clearance in response to an RA shall promptly return to the terms of that instruction or clearance when the conflict is resolved.

CAUTIONS

- ACAS is designed as a tool to improve situational awareness and as a last resort measure to avoid mid-air collision when other safety precautions fail.
- An intimate understanding of each specific system is required in order to accurately interpret the displayed information.
- The visual information displayed to the pilot is limited and can be misleading, caution is advised.

Reduced Vertical Separation Minima

- **Reduced Vertical Separation Minima (RVSM)** reduces the vertical separation between flight level (FL) 290–410 from 2000 ft to 1000 ft and makes six additional FL's available for operation.



General Flight Planning Consideration.

- Dispatcher and flight crew should pay particular attention to RVSM related items during flight planning process. Among these items are the following.
- Is the aircraft approved for RVSM operations.
- Are MEL items affecting RVSM operations.
- Has the flight plan correctly annotated in item 10 to show RVSM approval 'W'

ATS Flight Plan

- The ATS plan should indicate the appropriate RVSM level and contain the letter 'W' in field 10 to advise ATC the aircraft is RVSM capable.

Example:

(FPL-GFA071-IS
-A343/H-SRIXWEY/S
-OBBI0800
-N0470F320 DCT BAH A791 HIL F340 T540 ENABI UL550
NWB A791 MENLI A411 CVO DCT
-HECA0239 HESH HEGN
-EET/OEJD0044 HECC0207 REG/A4OLI SEL/GMBQ COM/TCAS)
-E/0506 P/TBN R/VE S/M J/LF C/FRY R)

Required Equipment for RVSM

REQUIRED EQUIPMENT	QUANTITY
ADR 1	1
ADR 2	1
ATC TRANSPONDER	1
FLIGHT WARNING COMPUTER (FOR ALTITUDE ALERT FUNCTION)	1
AUTOPILOT	1
PFD Function (For Altitude Indication)	2
FCU (FOR ALTITUDE TARGET SELECTION AND OP CLB/OP DES MODE ENGAGEMENT)	1

Note: A320/A321 family

Benefits of RVSM

- **Benefits of RVSM :**

- Fuel saving by getting the most optimum flight level.
- Increase airspace capacity by having 6 more levels. between FL290-410.
- Reduce ATC workload .
- Reduce en-route delay.

Thank you.

